

41st Indian Scientific Expedition to Antarctica (41-ISEA)

Operational Meteorological and Glaciological Field Observations at Maitri Station

1.0 Geographic Location and Environmental Regime

2.0 Automated Weather Station (AWS) Monthly Observations

Month	Mean Temp (°C)	Min Temp (°C)	Mean RH (%)	Wind (m/s)	Pressure (hPa)
January					
February					
March					
April					
May					
June					
July					
August					

Source: NCPOR Indian Antarctic Research Data Centre calibrated sensors (WMO ID: 89514).

3.0 Lake Priyadarshini Glaciological Observations

Lake Priyadarshini is a freshwater perennially ice-covered proglacial lake situated in the central sector of the Schirmacher Oasis. The lake has an approximate water volume of 0.85 million cubic meters and a maximum measured depth of 28 meters. Core sampling conducted during 41-ISEA revealed winter ice cover thickness ranging from 1.8 to 2.4 meters.

Continuous water quality telemetry logged by the benthic probe showed stable dissolved oxygen levels of 11.4 mg/L and surface water conductivity of 142 micro-Siemens per centimeter.

4.0 Atmospheric Aerosols and Solar Radiation Measurements

Multi-wavelength radiometer (MWR) observations at Maitri station showed background aerosol optical depth (AOD) at 500 nm wavelength hovering between 0.025 and 0.042 during clean continental maritime wind episodes. These ultra-low aerosol loading baselines confirm the pristine nature of the East Antarctic atmosphere and provide invaluable reference data for global climate change simulations and satellite calibration.

Table 2: Seasonal Solar Radiation and Net Flux Balance

Season	Direct Solar (W/m²)	Diffuse Flux (W/m²)	Albedo (%)	Net Radiation (W/m²)
Austral Summer (Dec-Feb)	780.4	145.2	22.4	+142.6
Austral Autumn (Mar-May)	210.8	68.4	38.6	-34.1
Austral Winter (Jun-Aug)	0.0	0.0	78.2	-86.5
Austral Spring (Sep-Nov)	490.2	112.5	45.1	+48.9

5.0 Institutional Provenance and Scientific Endorsement

This dataset was gathered under the aegis of the National Centre for Polar and Ocean Research (NCPOR), Ministry of Earth Sciences, New Delhi. All instruments comply with World Meteorological Organization (WMO) standards and were calibrated pre-expedition and post-deployment.

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